Unit 2 Review · Five Stops

Code name _____ Period ____

Unit 2 Cells · LT 2.1–2.5 · Walk-through: words first, watch one, try one, check one.

Stop 1 Cell theory

LT 2.1

Words to own: cell theory (three big ideas about cells) · **hypothesis** (a testable prediction) · **evidence** (what you can observe or measure)

Watch me. The three ideas: all living things are made of cells · cells are the basic unit of life · new cells come from cells that already exist. Observation: one cell splits into two. That supports the third idea.

Try. Scientists have looked at thousands of plants and animals, and every one is made of cells. Which idea does that support?

Check. Which one is a hypothesis? A. Cells are the basic unit of life. B. If I give this plant more light, it will grow taller. C. All cells have DNA.

Stop 2 Two kinds of cells and their parts

LT 2.2

Words to own: prokaryote (no nucleus; bacteria) · **eukaryote** (has a nucleus) · **ribosome** (builds proteins) · **mitochondria** (release energy from food)

Watch me. A bacterium has no nucleus, so it is a prokaryote. Every cell, prokaryote or eukaryote, has DNA, ribosomes and a cell membrane. Those three go in the middle of the diagram.

Try. Write each word in the right box: **nucleus · ribosomes · bacteria · mitochondria · DNA · cell membrane**

Prokaryote only	Both	Eukaryote only

Check. Circle the two that plant cells have and animal cells do not: cell wall · nucleus · chloroplast · mitochondria

Stop 3 Levels of organization and homeostasis

LT 2.3

Words to own: tissue (a group of similar cells) · **organ** (tissues working together) · **homeostasis** (keeping the inside steady) · **stomata** (leaf openings)

Watch me. Heart muscle cell → heart muscle tissue → heart → circulatory system → you. After a run, your heart beats fast, then slows back to normal. Going back to normal is homeostasis.

Try. Fill in the levels for a plant: leaf cell → _____ → leaf → _____ → bean plant

Try. On a hot afternoon a plant closes its stomata. How does that help it keep homeostasis?

Check. Which one carries out every life job inside a single cell? A. a skin cell B. an amoeba C. a leaf

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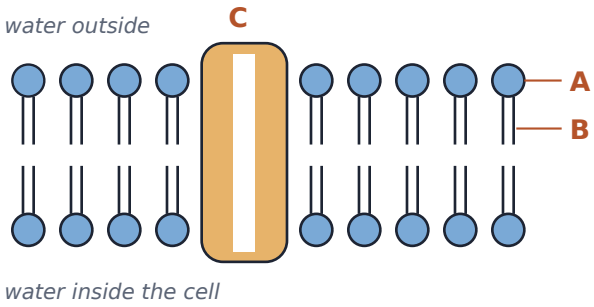
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Stop 4 The cell membrane

LT 2.4

Words to own: phospholipid · **hydrophilic** (water-loving head) · **hydrophobic** (water-fearing tail) · **selectively permeable** (lets some things through, not others)

Watch me. Water sits on both sides of the membrane, so the water-loving heads face out on both sides and the water-fearing tails hide in the middle. Proteins sit in the layer like doors.



Try. Name each part.

A _____

B _____

C _____ (what does it do?)

_____

Check. "Selectively permeable" means: A. nothing gets through B. everything gets through C. some things get through and others do not

Stop 5 Moving things across the membrane

LT 2.5 · a third of the test

Words to own: diffusion (high to low, no energy) · **osmosis** (diffusion of water) · **active transport** (low to high, uses energy) · **hypertonic** (more salt or sugar outside) · **hypotonic** (less outside)

Watch me. Water moves toward the side with more salt or sugar. A cell in salt water is in a **hypertonic** solution: water leaves and the cell shrinks. A cell in pure water is in a **hypotonic** solution: water rushes in and the cell swells.

Try. Draw an arrow for the water and say what happens to each red blood cell.



pure water
swells · same · shrinks

0.9% salt (same as the cell)
swells · same · shrinks

10% salt
swells · same · shrinks

Try. Left side 2% sugar, right side 8% sugar, and only water can cross. Water moves to the _____ side, and that side rises.

Check. Which one needs energy? A. osmosis B. diffusion C. a protein pump moving ions from low to high

Before the test: four moves

1. **Choose TWO?** Write T or F next to every choice first.
2. **Boxes and diagrams.** Every box is a point. Fill all of them, the middle too.
3. **Read the table out loud in your head.** Does it go up, peak, or drop?
4. **Osmosis.** Find the side with more salt or sugar. Water goes there.

Which stop felt shakiest? Circle it: 1 2 3 4 5

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